

CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 1 (T01) • Set A

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

A

Syllabus. Prerequisite material — probability, Python, linear algebra and descriptive statistics — together with Unit I (Lecture 1): what machine learning is, the three families, generalisation, cross-validation, data leakage and the Python stack; and Unit II (Lecture 2): loss functions, MSE, MAE, RMSE, Huber, cross-entropy, hinge and triplet loss.

Instructions. Closed book. Write your answers on this paper, in the space provided. A calculator is not needed — the arithmetic is deliberately short. Answer every question — there is no negative marking. In Part C a bare True/False without a reason scores half. In Parts D and E the marks are for the reasoning, not for the vocabulary; an answer that states what it cannot determine earns credit, an answer that asserts without a reason does not. Marks are shown against each part.

About the set letter. Every set of this quiz examines the same topics at the same level, for the same marks, against the same marking scheme. Your set reflects the *style* of question that suits how you work, and nothing else. It has no effect on your mark.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

A1. Let A be the event “at least one head” in two tosses of a fair coin. Then $P(A)$ equals

- (a) $3/4$ (b) $1/4$
(c) $1/2$ (d) 1

A2. `len(range(2, 10, 3))` evaluates to

- (a) 2 (b) 3
(c) 4 (d) an error

A3. A is 4×3 and B is 3×5 . The product AB has shape

- (a) 3×3 (b) 5×4
(c) 4×5 (d) undefined

A4. A model is fitted to labelled pairs (x_i, y_i) in which every y_i is a real number on a continuous scale. This is

- (a) supervised classification (b) unsupervised
(c) reinforcement (d) supervised regression

A5. Over constant predictions c , the quantity $\frac{1}{n} \sum_i |y_i - c|$ is minimised by the

- (a) median of the y_i (b) mean of the y_i
(c) mode of the y_i (d) largest y_i

Part B — Fill in the blanks

$4 \times 1 = 4$ marks. Write the missing word on the rule.

B1. For the values 2, 3, 4, 5, 96 the _____ exceeds the median.

B2. If $\text{RMSE}_{\text{train}} \ll \text{RMSE}_{\text{test}}$, the model is said to _____ the training data.

B3. If a step that learns from the data is fitted on the whole of X before X is split, test-row information reaches the model. This fault is called data _____.

B4. For one example, cross-entropy is the negative _____ of the probability assigned to the correct class.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one, then give a reason.

- C1.** A model achieving zero training error is guaranteed to perform well on unseen data. **TRUE** / **FALSE**

Reason: _____

- C2.** Model P reports $\text{MAE} = 1.0$ and model Q reports $\text{MSE} = 1.375$, both from the identical predictions; therefore P is the better model. **TRUE** / **FALSE**

Reason: _____

- C3.** For a point with $y = +1$ and score $f(x) = 3$, the hinge loss $\max(0, 1 - yf(x))$ is exactly zero, so that point does not push the boundary during training. **TRUE** / **FALSE**

Reason: _____

Part D — Short answer $2 \text{ questions} \times 2 = 4$ marks. Show how you decide, not just what you decide.

- D1.** For $y = (10, 10, 10)$ and $\hat{y} = (10, 13, 7)$: **(a)** compute the MSE and the MAE, showing your working; **(b)** state which of the two is more affected by the largest single residual, and why.

Part D — Short answer, continued

- D2.** A five-fold cross-validation of one model returns the fold scores (0.95, 0.93, 0.97, 0.62, 0.96). **(a)** Explain why reporting only their mean would be inadequate. **(b)** State what you would investigate first, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

- E1.** A colleague reports a test accuracy of 0.97 from the procedure below, on a dataset with $n = 100$ rows and $p = 5000$ columns. A domain expert says the honest figure should be close to chance.

Step 1. Load $X \in \mathbb{R}^{100 \times 5000}$ and $y \in \{0, 1\}^{100}$.

Step 2. Score all 5000 columns against y ; keep the 20 most strongly associated.

Step 3. Split the 100 rows into a training part and a test part.

Step 4. Fit a classifier on the training part.

Step 5. Report accuracy on the test part.

Answer all three parts overleaf. Any diagnosis that fits the evidence and is argued for earns full credit — naming the “expected” fault is not what is marked.

- (i) Name the most likely cause of the reported figure, and explain *why* it inflates it. [2]
- (ii) State one concrete check that would confirm or rule out your diagnosis. [1]
- (iii) Say what you would expect to see if your diagnosis turned out to be wrong. [1]

Part E — Reasoning, answer space Label each part (i), (ii), (iii). A sketch, a table or a list is as acceptable as prose.

Continue on a continuation sheet if you need more room; label anything you write there with its question number.

CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 1 (T01) • Set B

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

B

Syllabus. Prerequisite material — probability, Python, linear algebra and descriptive statistics — together with Unit I (Lecture 1): what machine learning is, the three families, generalisation, cross-validation, data leakage and the Python stack; and Unit II (Lecture 2): loss functions, MSE, MAE, RMSE, Huber, cross-entropy, hinge and triplet loss.

Instructions. Closed book. Write your answers on this paper, in the space provided. A calculator is not needed — the arithmetic is deliberately short. Answer every question — there is no negative marking. In Part C a bare True/False without a reason scores half. In Parts D and E the marks are for the reasoning, not for the vocabulary; an answer that states what it cannot determine earns credit, an answer that asserts without a reason does not. Marks are shown against each part.

About the set letter. Every set of this quiz examines the same topics at the same level, for the same marks, against the same marking scheme. Your set reflects the *style* of question that suits how you work, and nothing else. It has no effect on your mark.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

- A1.** The four equally likely outcomes of two fair coin tosses are HH, HT, TH, TT. The proportion containing at least one head is
(a) $1/4$ (b) $3/4$
(c) $1/2$ (d) 1
- A2.** For `lst = [10, 20, 30, 40]`, the expression `len(lst[1:3])` evaluates to
(a) 1 (b) 3
(c) 2 (d) 4
- A3.** An array of shape (4, 3) is matrix-multiplied by an array of shape (3, 5). The result has shape
(a) (3, 3) (b) (5, 4)
(c) undefined (d) (4, 5)
- A4.** A table has the columns *floor area*, *bedrooms* and *sale price in lakh*. Predicting the third from the first two is
(a) supervised regression (b) supervised classification
(c) unsupervised (d) reinforcement
- A5.** A model's four residuals are -1 , $+1$, $+1$, $+12$. The loss least disturbed by the value $+12$ is
(a) MSE (b) RMSE
(c) MAE (d) all equally

Part B — Fill in the blanks

4 $\times$ 1 = 4 marks. Write the missing word on the rule.

- B1.** For the readings 2, 3, 4, 5, 96 the mean is 22 and the median is 4; the _____ is the summary dragged upward by the single large reading.
- B2.** A model records a training RMSE of 0.4 and a test RMSE of 3.9. It is said to _____ the training data.
- B3.** A step that learns from the data is fitted on all 500 rows before the 400/100 split, so the test rows contribute to what it learned. This fault is called data _____.
- B4.** A model gives probability 0.7 to the correct class of one example; that example's cross-entropy is the negative _____ of 0.7.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one, then give a reason.

- C1.** A results table showing a training accuracy of 1.00 is sufficient evidence that the model will do well on new data.

TRUE / FALSEReason: _____

- C2.** A table reports $\text{MAE} = 1.0$ for model P and $\text{MSE} = 1.375$ for model Q , computed from the identical predictions. P is therefore the better model.

TRUE / FALSEReason: _____

- C3.** A table row records $y = +1$, $f(x) = 3$ and hinge loss $\max(0, 1 - yf(x)) = 0$; that zero means the point does not push the boundary during training.

TRUE / FALSEReason: _____

Part D — Short answer2 questions $\times$ 2 = 4 marks. Show how you decide, not just what you decide.

- D1.** A table records three true values and the three predictions made for them: $y = (6, 6, 6)$ and $\hat{y} = (5, 7, 10)$. **(a)** Compute the MSE and the MAE, showing your working. **(b)** State which of the two is more affected by the largest single residual, and why.

Part D — Short answer, continued

- D2.** A five-fold cross-validation of one model returns the fold scores (0.88, 0.91, 0.90, 0.55, 0.89). **(a)** Explain why reporting only their mean would be inadequate. **(b)** State what you would investigate first, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

- E1.** The table below records a colleague’s experiment. The domain expert consulted afterwards says the honest accuracy should be close to chance.

Rows in the dataset	100
Columns in the dataset	5000
Columns kept, scored against y over <i>all</i> 100 rows	20
Split performed	<i>after</i> those columns were kept
Reported test accuracy	0.97

Answer all three parts overleaf. Any diagnosis that fits the evidence and is argued for earns full credit — naming the “expected” fault is not what is marked.

- (i) Name the most likely cause of the reported figure, and explain *why* it inflates it. **[2]**
- (ii) State one concrete check that would confirm or rule out your diagnosis. **[1]**
- (iii) Say what you would expect to see if your diagnosis turned out to be wrong. **[1]**

Part E — Reasoning, answer space Label each part (i), (ii), (iii). A sketch, a table or a list is as acceptable as prose.

Continue on a continuation sheet if you need more room; label anything you write there with its question number.

CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 1 (T01) • Set C

Duration 40 minutes • Maximum marks 20 • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

C

Syllabus. Prerequisite material — probability, Python, linear algebra and descriptive statistics — together with Unit I (Lecture 1): what machine learning is, the three families, generalisation, cross-validation, data leakage and the Python stack; and Unit II (Lecture 2): loss functions, MSE, MAE, RMSE, Huber, cross-entropy, hinge and triplet loss.

Instructions. Closed book. Write your answers on this paper, in the space provided. A calculator is not needed — the arithmetic is deliberately short. Answer every question — there is no negative marking. In Part C a bare True/False without a reason scores half. In Parts D and E the marks are for the reasoning, not for the vocabulary; an answer that states what it cannot determine earns credit, an answer that asserts without a reason does not. Marks are shown against each part.

About the set letter. Every set of this quiz examines the same topics at the same level, for the same marks, against the same marking scheme. Your set reflects the *style* of question that suits how you work, and nothing else. It has no effect on your mark.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

- A1.** A probability tree for two tosses of a fair coin has four leaves of equal weight. The combined weight of the leaves showing at least one head is
(a) $1/4$ (b) $1/2$
(c) $3/4$ (d) 1
- A2.** `len(set([1, 2, 2, 3]))` evaluates to
(a) 1 (b) 2
(c) 4 (d) 3
- A3.** One rectangle is drawn 4 rows tall and 3 columns wide, a second 3 rows tall and 5 columns wide. Multiplied in that order, the result rectangle is
(a) 4 by 5 (b) 3 by 3
(c) 5 by 4 (d) not defined
- A4.** A scatter plot of two features carries no colour coding at all, and the task is to draw three natural clumps around the points. This is
(a) supervised regression (b) unsupervised
(c) supervised classification (d) reinforcement
- A5.** Two loss curves are plotted against the error e . One is a smooth parabola through the origin; the other is a V with a sharp corner at the origin. The V is
(a) MSE (b) hinge loss
(c) cross-entropy (d) MAE

Part B — Fill in the blanks

4 $\times$ 1 = 4 marks. Write the missing word on the rule.

- B1.** A histogram of 2, 3, 4, 5, 96 has a long right tail. The vertical line marking the _____ sits far to the right of the line marking the median.
- B2.** On a learning curve the training error keeps falling while the validation error turns upward. From that turning point onward the model is said to _____.
- B3.** A pipeline diagram shows the box that learns from the data placed *before* the split box rather than inside the fold. The fault this creates is called data _____.
- B4.** A plot of $-\log p$ against p rises without bound as $p \rightarrow 0$. Cross-entropy for one example is exactly this curve: the negative _____ of the probability given to the correct class.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one, then give a reason.

- C1.** A learning curve whose training-error line reaches zero shows a model that will generalise well. **TRUE** / **FALSE**

Reason: _____

- C2.** Two bars on one chart, labelled $\text{MAE} = 1.0$ for model P and $\text{MSE} = 1.375$ for model Q and computed from identical predictions, show that P is the better model. **TRUE** / **FALSE**

Reason: _____

- C3.** On a plot of hinge loss against the margin $yf(x)$, the curve is flat at zero for every margin beyond 1; a point sitting at margin 3 therefore exerts no pull on the boundary. **TRUE** / **FALSE**

Reason: _____

Part D — Short answer $2 \text{ questions} \times 2 = 4$ marks. Show how you decide, not just what you decide.

- D1.** Three points are plotted against a horizontal line at $y = 8$. Their predicted heights are $\hat{y} = (6, 10, 13)$ while every true value is $y = 8$. **(a)** Compute the MSE and the MAE, showing your working. **(b)** State which of the two is more affected by the largest single residual, and why.

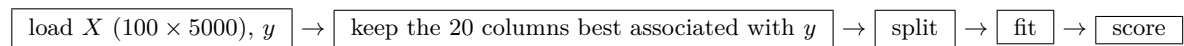
Part D — Short answer, continued

- D2.** A bar chart of five cross-validation fold scores shows four bars of similar height and one much shorter: (0.92, 0.90, 0.94, 0.61, 0.93). **(a)** Explain why reporting only their mean would be inadequate. **(b)** State what you would investigate first, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

- E1.** A colleague's plot shows the reported test accuracy sitting at 0.97, *above* every one of the training-fold accuracies drawn beside it — an ordering that should not occur. The dataset is 100 rows by 5000 columns. Reading the pipeline diagram from left to right, the boxes are



Answer all three parts overleaf. Any diagnosis that fits the evidence and is argued for earns full credit — naming the “expected” fault is not what is marked.

- (i) Name the most likely cause of the reported figure, and explain *why* it inflates it. [2]
- (ii) State one concrete check that would confirm or rule out your diagnosis. [1]
- (iii) Say what you would expect to see if your diagnosis turned out to be wrong. [1]

Part E — Reasoning, answer space Label each part (i), (ii), (iii). A sketch, a table or a list is as acceptable as prose.

Continue on a continuation sheet if you need more room; label anything you write there with its question number.

CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 1 (T01) • Set D

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

D

Syllabus. Prerequisite material — probability, Python, linear algebra and descriptive statistics — together with Unit I (Lecture 1): what machine learning is, the three families, generalisation, cross-validation, data leakage and the Python stack; and Unit II (Lecture 2): loss functions, MSE, MAE, RMSE, Huber, cross-entropy, hinge and triplet loss.

Instructions. Closed book. Write your answers on this paper, in the space provided. A calculator is not needed — the arithmetic is deliberately short. Answer every question — there is no negative marking. In Part C a bare True/False without a reason scores half. In Parts D and E the marks are for the reasoning, not for the vocabulary; an answer that states what it cannot determine earns credit, an answer that asserts without a reason does not. Marks are shown against each part.

About the set letter. Every set of this quiz examines the same topics at the same level, for the same marks, against the same marking scheme. Your set reflects the *style* of question that suits how you work, and nothing else. It has no effect on your mark.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

- A1.** Two independent inspections of a component each pass with probability $1/2$. The probability that at least one passes is
(a) $1/4$ (b) $1/2$
(c) 1 (d) $3/4$
- A2.** A shop's stock is written as `{'red': 1, 'blue': 2, 'red': 3}`. Its `len` is
(a) 2 (b) 1
(c) 3 (d) an error
- A3.** A clinic records 8 measurements for each of 200 patients. To score a patient with one weight per measurement, the weight vector must have length
(a) 200 (b) 8
(c) 1600 (d) any length
- A4.** A bank holds five years of loans, each marked *repaid* or *defaulted*, and wants a verdict for a new applicant. This is
(a) supervised regression (b) unsupervised
(c) supervised classification (d) reinforcement
- A5.** A temperature sensor occasionally reports impossible values that cannot be removed before training. The loss whose fitted constant is least disturbed by them is
(a) MSE (b) MAE
(c) RMSE (d) squared hinge

Part B — Fill in the blanks

4 $\times$ 1 = 4 marks. Write the missing word on the rule.

- B1.** A ward records stays of 2, 3, 4, 5 and 96 days. The _____ length of stay is the one pulled upward by the single long stay.
- B2.** A churn model is right about almost every past customer and wrong about most new ones. It is said to _____ the training data.
- B3.** An analyst lets a step that learns from the data see the full customer table, then splits it into train and test parts. The fault is called data _____.
- B4.** For one email the spam filter gives probability 0.7 to the class the email actually belongs to. That email's cross-entropy is the negative _____ of 0.7.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one, then give a reason.

- C1.** A fraud model that flags every past fraud in the bank's own records correctly is ready to be deployed on next month's transactions. **TRUE / FALSE**

Reason: _____

- C2.** A vendor's model reports $\text{MAE} = 1.0$ and yours reports $\text{MSE} = 1.375$ on the identical predictions; the vendor's model is therefore better. **TRUE / FALSE**

Reason: _____

- C3.** A support-vector classifier scores a correctly labelled application at $f(x) = 3$ with $y = +1$, so its hinge loss is zero and removing that application from the training set would not move the boundary. **TRUE / FALSE**

Reason: _____

Part D — Short answer $2 \text{ questions} \times 2 = 4$ marks. Show how you decide, not just what you decide.

- D1.** A delivery service quotes 20 minutes for three consecutive orders, so $y = (20, 20, 20)$. The model's predictions were $\hat{y} = (19, 24, 16)$. **(a)** Compute the MSE and the MAE, showing your working. **(b)** State which of the two is more affected by the largest single residual, and why.

Part D — Short answer, continued

- D2.** A five-fold cross-validation of a churn model returns the fold scores (0.86, 0.84, 0.88, 0.52, 0.87). **(a)** Explain why reporting only their mean would be inadequate. **(b)** State what you would investigate first, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

- E1.** An intern presents a churn model at the weekly meeting and reports a test accuracy of 0.97. The dataset holds 100 customers and 5000 recorded attributes per customer. The team's own domain expert says the honest figure should be close to chance, because most of those attributes are meaningless. Asked how the result was produced, the intern says

“I looked at all 5000 attributes across all 100 customers, kept the 20 that tracked churn most closely, then split the customers into a training set and a test set, trained on the training set, and scored on the test set.”

Answer all three parts overleaf. Any diagnosis that fits the evidence and is argued for earns full credit — naming the “expected” fault is not what is marked.

- (i) Name the most likely cause of the reported figure, and explain *why* it inflates it. [2]
- (ii) State one concrete check that would confirm or rule out your diagnosis. [1]
- (iii) Say what you would expect to see if your diagnosis turned out to be wrong. [1]

Part E — Reasoning, answer space Label each part (i), (ii), (iii). A sketch, a table or a list is as acceptable as prose.

Continue on a continuation sheet if you need more room; label anything you write there with its question number.

CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 1 (T01) • Set E

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

E

Syllabus. Prerequisite material — probability, Python, linear algebra and descriptive statistics — together with Unit I (Lecture 1): what machine learning is, the three families, generalisation, cross-validation, data leakage and the Python stack; and Unit II (Lecture 2): loss functions, MSE, MAE, RMSE, Huber, cross-entropy, hinge and triplet loss.

Instructions. Closed book. Write your answers on this paper, in the space provided. A calculator is not needed — the arithmetic is deliberately short. Answer every question — there is no negative marking. In Part C a bare True/False without a reason scores half. In Parts D and E the marks are for the reasoning, not for the vocabulary; an answer that states what it cannot determine earns credit, an answer that asserts without a reason does not. Marks are shown against each part.

About the set letter. Every set of this quiz examines the same topics at the same level, for the same marks, against the same marking scheme. Your set reflects the *style* of question that suits how you work, and nothing else. It has no effect on your mark.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

A1. After `o = list(product('HT', repeat=2))`, the value of `sum(1 for x in o if 'H' in x) / len(o)` is

- (a) 0.75 (b) 0.25
(c) 0.5 (d) 1.0

A2. After `xs = [i for i in range(5) if i % 2 == 0]`, `len(xs)` is

- (a) 2 (b) 4
(c) 3 (d) 5

A3. `np.zeros((4, 3)) @ np.ones((3, 5))` has shape

- (a) (3, 3) (b) (4, 5)
(c) (5, 4) (d) it raises

A4. `KMeans(n_clusters=3).fit(X)` is called and no `y` is passed anywhere. The task is

- (a) supervised regression (b) supervised classification
(c) reinforcement (d) unsupervised

A5. `np.mean(np.abs(y - y_hat))` computes

- (a) MSE (b) RMSE
(c) MAE (d) cross-entropy

Part B — Fill in the blanks

4 $\times$ 1 = 4 marks. Write the missing word on the rule.

B1. `np.mean([2, 3, 4, 5, 96])` returns 22.0 and `np.median(...)` returns 4.0. The _____ is the one moved by the value 96.

B2. `model.score(X_train, y_train)` returns 0.99 and `model.score(X_test, y_test)` returns 0.61. The model is said to _____ the training data.

B3. A step that learns from the data is called on the whole of `X` before `train_test_split` is reached. The fault this creates is called data _____.

B4. For one example, `-np.log(p_correct)` is the cross-entropy: the negative _____ of the probability given to the correct class.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one, then give a reason.

C1. `model.score(X_train, y_train) == 1.0` is enough to conclude that the model will do well on unseen data.
TRUE / FALSE

Reason: _____

C2. From the identical arrays, `mean_absolute_error(...)` returns 1.0 for model P and `mean_squared_error(...)` returns 1.375 for model Q . Since $1.0 < 1.375$, P is the better model. **TRUE / FALSE**

Reason: _____

C3. For $y = 1$ and $f = 3.0$, `max(0.0, 1 - y * f)` evaluates to 0.0, so this point contributes nothing to the hinge objective and does not move the boundary. **TRUE / FALSE**

Reason: _____

Part D — Short answer $2 \text{ questions} \times 2 = 4$ marks. Show how you decide, not just what you decide.

D1. `y = np.array([12, 12, 12])` and `y_hat = np.array([14, 10, 20])`. (a) Compute the MSE and the MAE by hand, showing your working. (b) State which of the two is more affected by the largest single residual, and why.

Part D — Short answer, continued

D2. `cross_val_score(model, X, y, cv=5)` returns `array([0.94, 0.92, 0.96, 0.60, 0.95])`. **(a)** Explain why reporting only `.mean()` would be inadequate. **(b)** State what you would investigate first, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

E1. The function below is run on `X` of shape `(100, 5000)` and a binary `y`, and returns 0.97. A domain expert says the honest figure should be close to chance.

```
from sklearn.feature_selection import SelectKBest, f_classif
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier

def evaluate(X, y):
    X_sel = SelectKBest(f_classif, k=20).fit_transform(X, y)
    X_tr, X_te, y_tr, y_te = train_test_split(X_sel, y, test_size=0.25,
                                              random_state=0)
    model = KNeighborsClassifier(n_neighbors=3).fit(X_tr, y_tr)
    return model.score(X_te, y_te)
```

Answer all three parts overleaf. Any diagnosis that fits the evidence and is argued for earns full credit — naming the “expected” fault is not what is marked.

- (i) Name the most likely cause of the reported figure, and explain *why* it inflates it. [2]
- (ii) State one concrete check that would confirm or rule out your diagnosis. [1]
- (iii) Say what you would expect to see if your diagnosis turned out to be wrong. [1]

Part E — Reasoning, answer space Label each part (i), (ii), (iii). A sketch, a table or a list is as acceptable as prose.

Continue on a continuation sheet if you need more room; label anything you write there with its question number.

CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 1 (T01) • Set F

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

F

Syllabus. Prerequisite material — probability, Python, linear algebra and descriptive statistics — together with Unit I (Lecture 1): what machine learning is, the three families, generalisation, cross-validation, data leakage and the Python stack; and Unit II (Lecture 2): loss functions, MSE, MAE, RMSE, Huber, cross-entropy, hinge and triplet loss.

Instructions. Closed book. Write your answers on this paper, in the space provided. A calculator is not needed — the arithmetic is deliberately short. Answer every question — there is no negative marking. In Part C a bare True/False without a reason scores half. In Parts D and E the marks are for the reasoning, not for the vocabulary; an answer that states what it cannot determine earns credit, an answer that asserts without a reason does not. Marks are shown against each part.

About the set letter. Every set of this quiz examines the same topics at the same level, for the same marks, against the same marking scheme. Your set reflects the *style* of question that suits how you work, and nothing else. It has no effect on your mark.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

- A1.** A fair coin is tossed twice. The chance of getting at least one head is
 (a) one in four (b) three in four
 (c) one in two (d) certain
- A2.** In Python, splitting the word *machine* on the letter *h* produces a list. Its length is
 (a) 1 (b) 3
 (c) 7 (d) 2
- A3.** A rectangular table of numbers with four rows and three columns is multiplied by a second table with three rows and five columns. The result has
 (a) three rows, three columns (b) five rows, four columns
 (c) four rows, five columns (d) no defined result
- A4.** A newsroom wants its articles grouped by topic, but nobody has ever labelled an article with a topic. This is
 (a) unsupervised (b) supervised regression
 (c) supervised classification (d) reinforcement
- A5.** Two ways of scoring the same errors: one squares each error before averaging, the other takes the size of each before averaging. The second is less disturbed by a single wild reading. It is
 (a) MSE (b) RMSE
 (c) hinge loss (d) MAE

Part B — Fill in the blanks

4 $\times$ 1 = 4 marks. Write the missing word on the rule.

- B1.** When one value in a small set of readings is far larger than the rest, the _____ is pulled towards it while the median stays put.
- B2.** A model that has learned the particular examples it was shown rather than the pattern behind them is said to _____.
- B3.** Using, at training time, information that would not have been available at prediction time is called data _____.
- B4.** For a single example, cross-entropy is the negative _____ of the probability the model gave to the correct class.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one, then give a reason.

- C1.** Getting every training example right proves that a model has learned the underlying pattern. **TRUE** / **FALSE**

Reason: _____

- C2.** One model's mean absolute error is a smaller number than another model's mean squared error on the same predictions, so the first model is the better one. **TRUE** / **FALSE**

Reason: _____

- C3.** A correctly classified point that already sits comfortably beyond the margin costs a hinge-loss model nothing, and therefore has no say in where the boundary is placed. **TRUE** / **FALSE**

Reason: _____

Part D — Short answer $2 \text{ questions} \times 2 = 4$ marks. Show how you decide, not just what you decide.

- D1.** Three quantities were each truly fifteen, and the model predicted twelve, eighteen and twenty-one for them in that order. **(a)** Compute the mean squared error and the mean absolute error, showing your working. **(b)** State which of the two is more affected by the largest single residual, and why.

Part D — Short answer, continued

- D2.** Splitting the data five ways and scoring each part in turn gave five numbers: 0.90, 0.88, 0.92, 0.57 and 0.91. **(a)** Explain why reporting only their average would be inadequate. **(b)** State what you would investigate first, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

- E1.** A colleague describes what she did, and reports an accuracy of ninety-seven per cent. The data she used holds one hundred records, with five thousand recorded quantities per record. A domain expert says the honest figure should be close to chance.

“I went through all five thousand quantities, using every one of the hundred records, and kept the twenty that lined up best with the answer I was trying to predict. Then I set part of the data aside as a test set, trained on the rest, and measured how well it did on the part I had set aside.”

Answer all three parts overleaf. Any diagnosis that fits the evidence and is argued for earns full credit — naming the “expected” fault is not what is marked.

- (i) Name the most likely cause of the reported figure, and explain *why* it inflates it. [2]
- (ii) State one concrete check that would confirm or rule out your diagnosis. [1]
- (iii) Say what you would expect to see if your diagnosis turned out to be wrong. [1]

Part E — Reasoning, answer space Label each part (i), (ii), (iii). A sketch, a table or a list is as acceptable as prose.

Continue on a continuation sheet if you need more room; label anything you write there with its question number.

CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 1 (T01) • Set G

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

G

Syllabus. Prerequisite material — probability, Python, linear algebra and descriptive statistics — together with Unit I (Lecture 1): what machine learning is, the three families, generalisation, cross-validation, data leakage and the Python stack; and Unit II (Lecture 2): loss functions, MSE, MAE, RMSE, Huber, cross-entropy, hinge and triplet loss.

Instructions. Closed book. Write your answers on this paper, in the space provided. A calculator is not needed — the arithmetic is deliberately short. Answer every question — there is no negative marking. In Part C a bare True/False without a reason scores half. In Parts D and E the marks are for the reasoning, not for the vocabulary; an answer that states what it cannot determine earns credit, an answer that asserts without a reason does not. Marks are shown against each part.

About the set letter. Every set of this quiz examines the same topics at the same level, for the same marks, against the same marking scheme. Your set reflects the *style* of question that suits how you work, and nothing else. It has no effect on your mark.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

- A1.** For two tosses of a fair coin, compare $P(\text{at least one head})$ with $P(\text{exactly one head})$
 (a) the second is larger (b) they are equal
 (c) the first is larger (d) cannot be decided
- A2.** Compare `len([1, 2, 3][:2])` with `len([1, 2] + [3])`
 (a) the second is larger (b) the first is larger
 (c) they are equal (d) one raises an error
- A3.** A is 4×3 and B is 3×5 . Comparing the two products
 (a) only BA is defined (b) both, same shape
 (c) both, different shapes (d) only AB is defined
- A4.** Compare (i) predicting a house's sale price with (ii) deciding whether an email is spam
 (a) (i) classification, (ii) regression (b) (i) regression, (ii) classification
 (c) both regression (d) both unsupervised
- A5.** Compare MSE and MAE on a set of residuals containing one very large value. The one moved further by that value is
 (a) MSE (b) MAE
 (c) neither, they move equally (d) it depends on the sign

Part B — Fill in the blanks

4 $\times$ 1 = 4 marks. Write the missing word on the rule.

- B1.** Compare the mean and median of 2, 3, 4, 5, 96; the larger of the two is the _____.
- B2.** Compare a model's error on the data it was fitted to with its error on unseen data. When the second is far worse, the model is said to _____.
- B3.** Compare fitting a step that learns from the data *inside* each fold with fitting it once on the whole dataset. The second choice produces data _____.
- B4.** Compare accuracy with cross-entropy: accuracy merely counts, whereas cross-entropy is the negative _____ of the probability given to the correct class.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one, then give a reason.

- C1.** Between two models, the one with the higher training accuracy is the one that will do better on new data.

TRUE / FALSEReason: _____

- C2.** Between model P ($\text{MAE} = 1.0$) and model Q ($\text{MSE} = 1.375$) on identical predictions, P is the better model.

TRUE / FALSEReason: _____

- C3.** Between a point at margin $yf(x) = 3$ and a point at margin $yf(x) = 0.5$, only the second contributes to the hinge objective; the first costs zero and does not move the boundary. **TRUE / FALSE**

Reason: _____

Part D — Short answer $2 \text{ questions} \times 2 = 4$ marks. Show how you decide, not just what you decide.

- D1.** Compare three predictions $\hat{y} = (9, 12, 3)$ against three true values $y = (9, 9, 9)$. **(a)** Compute the MSE and the MAE, showing your working. **(b)** State which of the two is more affected by the largest single residual, and why.

Part D — Short answer, continued

- D2.** Compare the five fold scores returned by a five-fold cross-validation: (0.93, 0.91, 0.95, 0.59, 0.94). **(a)** Explain why reporting only their mean would be inadequate. **(b)** State what you would investigate first, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

- E1.** Two colleagues run the same dataset — 100 rows, 5000 columns — through the procedures below. The procedures use the identical four steps in a different order, and report very different figures. The domain expert says the honest figure should be close to chance.

Procedure A — reports 0.97	Procedure B — reports 0.51
1. Keep the 20 columns best associated with y .	1. Split into train and test.
2. Split into train and test.	2. Keep the 20 columns best associated with y .
3. Fit a classifier.	3. Fit a classifier.
4. Report accuracy on the test part.	4. Report accuracy on the test part.

Answer all three parts overleaf. Any diagnosis that fits the evidence and is argued for earns full credit — naming the “expected” fault is not what is marked.

- (i) Name the most likely cause of the reported figure, and explain *why* it inflates it. [2]
- (ii) State one concrete check that would confirm or rule out your diagnosis. [1]
- (iii) Say what you would expect to see if your diagnosis turned out to be wrong. [1]

Part E — Reasoning, answer space Label each part (i), (ii), (iii). A sketch, a table or a list is as acceptable as prose.

Continue on a continuation sheet if you need more room; label anything you write there with its question number.

CSAI2017P — Applied Machine Learning (Theory)

Theory Quiz 1 (T01) • Set H

Duration **40 minutes** • Maximum marks **20** • 15 questions on four pages

Name _____ SAP ID _____ Batch _____

Date _____ Signature _____ Set

H

Syllabus. Prerequisite material — probability, Python, linear algebra and descriptive statistics — together with Unit I (Lecture 1): what machine learning is, the three families, generalisation, cross-validation, data leakage and the Python stack; and Unit II (Lecture 2): loss functions, MSE, MAE, RMSE, Huber, cross-entropy, hinge and triplet loss.

Instructions. Closed book. Write your answers on this paper, in the space provided. A calculator is not needed — the arithmetic is deliberately short. Answer every question — there is no negative marking. In Part C a bare True/False without a reason scores half. In Parts D and E the marks are for the reasoning, not for the vocabulary; an answer that states what it cannot determine earns credit, an answer that asserts without a reason does not. Marks are shown against each part.

About the set letter. Every set of this quiz examines the same topics at the same level, for the same marks, against the same marking scheme. Your set reflects the *style* of question that suits how you work, and nothing else. It has no effect on your mark.

Part A — Multiple choice

5 questions $\times$ 1 = 5 marks. Circle one letter.

- A1.** For one toss of a fair coin, $P(\text{head}) = 1/2$. For two tosses, $P(\text{at least one head})$ is
 (a) $1/4$ (b) $1/2$
 (c) 1 (d) $3/4$
- A2.** $\text{len}([1, 2, 3][0:2])$ is 2. Then $\text{len}([5, 6, 7, 8][1:4])$ is
 (a) 2 (b) 3
 (c) 4 (d) an error
- A3.** A 2×3 times a 3×4 gives a 2×4 . Then a 4×3 times a 3×5 gives
 (a) 4×5 (b) 3×3
 (c) 5×4 (d) undefined
- A4.** Predicting a continuous quantity from labelled examples is supervised regression. Sorting unlabelled photographs into groups nobody has named is
 (a) supervised regression (b) supervised classification
 (c) unsupervised (d) reinforcement
- A5.** The constant minimising $\sum_i (y_i - c)^2$ is the mean. The constant minimising $\sum_i |y_i - c|$ is the
 (a) mean (b) median
 (c) mode (d) largest value

Part B — Fill in the blanks

$4 \times 1 = 4$ marks. Write the missing word on the rule.

- B1.** For 1, 2, 3 the mean is 2 and so is the median. For 2, 3, 4, 5, 96 they part company, and the larger of the two is the _____.
- B2.** A model with training error 0.40 and test error 0.45 is generalising. A model with training error 0.40 and test error 3.90 is instead said to _____.
- B3.** A step that learns from the data, fitted inside each fold, is safe. The same step fitted on the whole dataset before the split is data _____.
- B4.** The probability given to the correct class was 0.9, so the cross-entropy was $-\log 0.9$. In general it is the negative _____ of that probability.

Question	A1	A2	A3	A4	A5
Answer					

Copy your circled Part A letters into this grid. The grid is what is marked.

Part C — True or false $3 \times 1 = 3$ marks. Circle one, then give a reason.

- C1.** *A model with zero training error and zero test error has generalised.* Therefore a model with zero training error has generalised. **TRUE / FALSE**

Reason: _____

- C2.** *Comparing two models by the same metric on the same held-out data is valid.* Comparing model P 's MAE of 1.0 with model Q 's MSE of 1.375 is therefore also valid, and shows P to be better. **TRUE / FALSE**

Reason: _____

- C3.** *For $y = +1$ and $f(x) = 0.4$ the hinge loss is 0.6, and the point pulls on the boundary.* For $y = +1$ and $f(x) = 3$ the hinge loss is 0, and the point does not pull on the boundary. **TRUE / FALSE**

Reason: _____

Part D — Short answer2 questions $\times$ 2 = 4 marks. Show how you decide, not just what you decide.

- D1.** *For $y = (4, 4)$ and $\hat{y} = (3, 5)$ the residuals are -1 and $+1$, so $\text{MSE} = 1$ and $\text{MAE} = 1$.* Now take $y = (30, 30, 30)$ and $\hat{y} = (28, 35, 25)$. **(a)** Compute the MSE and the MAE, showing your working. **(b)** State which of the two is more affected by the largest single residual, and why.

Part D — Short answer, continued

D2. *Fold scores (0.90, 0.91, 0.89) are tightly grouped, and their mean is a fair summary.* A five-fold run now returns (0.89, 0.87, 0.91, 0.54, 0.90). **(a)** Explain why reporting only their mean would be inadequate. **(b)** State what you would investigate first, and why.

Part E — Reasoning

1 question $\times$ 4 = 4 marks. You are marked on the reasoning you make visible.

E1. *Worked analogue.* A student tunes a threshold by trying twenty values and reporting the best test score. The reported figure is optimistic, because the test set has been consulted twenty times; the check is to re-run with the tuning done inside the training data only, and the score should fall.

Now consider this case. A colleague reports a test accuracy of 0.97 on a dataset of 100 rows and 5000 columns, where the domain expert says the honest figure should be close to chance. The colleague scored all 5000 columns against y using all 100 rows, kept the best 20, then split into train and test, fitted a classifier and reported the test accuracy.

Answer all three parts overleaf. Any diagnosis that fits the evidence and is argued for earns full credit — naming the “expected” fault is not what is marked.

- (i) Name the most likely cause of the reported figure, and explain *why* it inflates it. [2]
- (ii) State one concrete check that would confirm or rule out your diagnosis. [1]
- (iii) Say what you would expect to see if your diagnosis turned out to be wrong. [1]

Part E — Reasoning, answer space Label each part (i), (ii), (iii). A sketch, a table or a list is as acceptable as prose.

Continue on a continuation sheet if you need more room; label anything you write there with its question number.